MILITARY SPECIFICATION

SUCTION APPARATUS, SURGICAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a high volume, low pressure pleural cavity pump, suitable for maintaining negativity in the pleural space by rapid removal of air during post-operative procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-C-77	Caster, Institutional Duty.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall.

Military

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-5541	Chemical Conservation Coatings on Aluminum and Aluminum Alloys.
MIL-M-7298	Manual, Technical, Commercial Equipment.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE INC. (ANSI)

C 73.11	Plugs and Receptacles - General Purpose. 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to American National Standards Institute Inc., 1430 Broadway, New York, New York 10018.)

ASSOCIATION FOR THE ADVANCEMENT OF MEDICAL INSTRUMENTATION (AAMI)

AAMI Safety Standard for Electromedical Apparatus: Safe Current Limits.

(Application for copies should be addressed to Association for the Advancement of Medical Instrumentation, 1500 Wilson Blvd, Suite 417, Arlington, VA 22209)

3. REQUIREMENTS

3.1 Materials. Materials used in the construction of the suction apparatus shall be of the highest quality throughout and suitable in all respects for the intended purpose. Corrosion-resisting steel used in construction of the unit shall be AISI 302 or 304. Other materials shall be as specified herein. Material gages specified shall be considered the minimum acceptable gages. Commercial tolerances shall apply to gage thickness.

3.2 Design and construction. The suction apparatus shall be a mobile, high-volume, low-pressure type for continuous bedside use in wards and recovery rooms. The unit shall be constructed in conformance to type I or type

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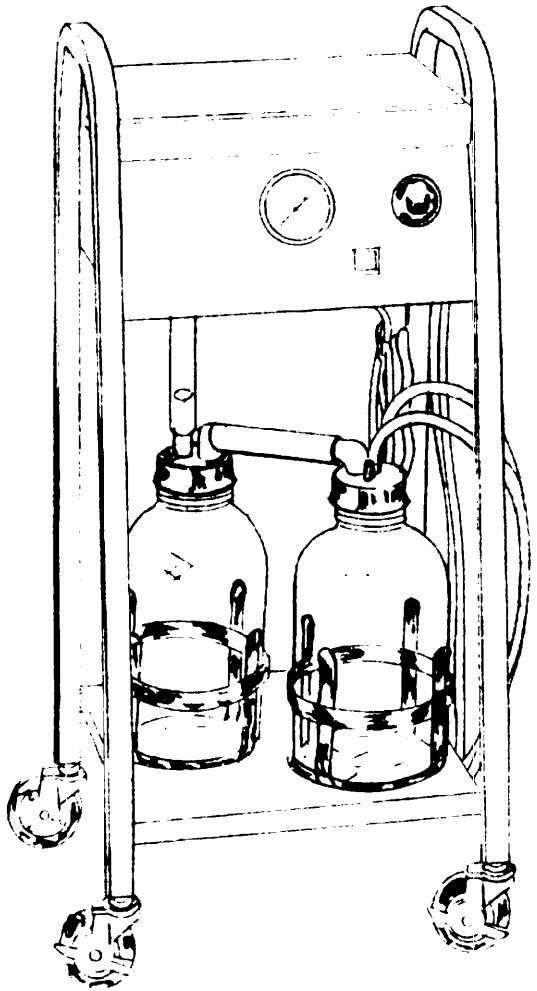
II of Figure I. The unit shall consist of a cabinet containing an electric motor and vacuum pump assembly, a variable transformer, on-off switch, vacuum gage, indicator light, night light, and a three-conductor power cord suitably interconnected. The cabinet shall be mounted on a mobile stand equipped with a tray or shelf capable of holding three collection bottles. Two trap bottles shall be provided complete with caps, fittings, and tubing necessary to interconnect the bottles to themselves, the pump and the patient. The unit shall be approximately 33 inches high by 20 inches wide by 18 inches deep. The weight of the unit shall not exceed 36 pounds.

3.2.1 Cabinet. The cabinet shall be fabricated of cold rolled steel, or aluminum and corrosion-resisting steel. If cold rolled steel is used, the cabinet front panel shall be 0.049 inch thick (No. 18 United States Standard (U.S.S.) gage) sheet and the cabinet body shall be 0.029 inch thick (No. 22 U.S.S. gage) sheet. If aluminum and corrosion-resisting steel are used, the cabinet body shall be fabricated of 0.040 inch thick (No. 18 Brown and Sharpe (B and S) gage) aluminum sheet and the cabinet top shall be fabricated of 0.025 inch thick (No. 24 U.S.S. gage) corrosion-resisting steel sheet. The on-off switch, the variable transformer, vacuum gage, and indicator light shall be mounted on the front of the cabinet. The bottom of the cabinet shall contain a three prong receptacle, that shall accept a plug-in night light, and all necessary mounting bracket holes. Reinforcing plates shall be employed, if necessary, to stiffen the unit. The cabinet shall be sturdy and easily accessible for repair. A caution notice to the effect that the unit is not suitable for use in hazardous atmospheres, shall be printed on the cabinet front.

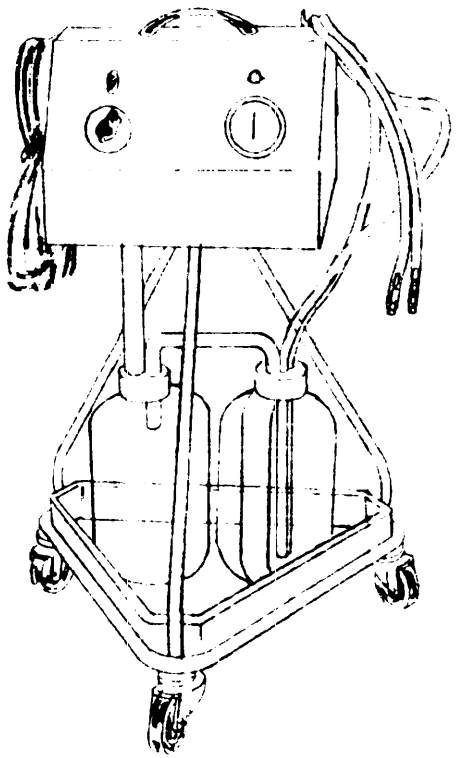
3.2.2 Mobile stand. The mobile stand shall be sturdily constructed in the configuration of Type I or Type II of Figure 1. The stand frame of the Type I unit shall be constructed of 5/8 inch O.D. corrosion-resisting steel tubing with a nominal wall thickness of 0.062 inch. Commercial tolerances shall apply to all gages and thicknesses. The stand frame of the Type II unit shall be constructed of 1 inch square aluminum tubing 0.065 inch minimum wall thickness (No. 16 B and S Gage). The Type I stand shall be equipped with a triangular 2 inch deep plastic tray capable of holding three collection bottles. The Type II stand shall be equipped with an aluminum shelf, with two corrosion-resisting steel bottle holders, capable of accommodating an additional bottle if necessary. The stand shall be mounted on swivel casters with 3-inch diameter rubber wheels. The cabinet shall be securely fastened to the stand. Casters shall conform to Type II, Class A, style B, mounting A of FF-C-77. National Sanitation Foundation (NSF) requirements in FF-C-77 shall apply. Two of the casters on each unit shall have wheel-brakes. On the Type II unit brakes shall be on the front casters.

3.2.3 Motor and pump assembly. A high-volume, low-pressure vacuum pump and motor assembly shall be securely mounted in the cabinet. The motor shall be integrally connected to the pump. The motor shall be of the series universal

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TYPE II



TYPE I

SUCTION APPARATUS, SURGICAL

FIGURE I

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type with permanently lubricated bearings and shall be suitable for operation on 120 volt, 60 Hertz (Hz) electric power. An intake dome and gasket shall be completely sealed to the pump and gasket to prevent air leaks at these points. The dome shall be connected by means of fittings and tubing to the collection bottles and the vacuum gage. An additional tubing connection from the intake dome vented to the atmosphere at the bottom of the cabinet shall be providied. The assembly shall be securely held in place by a shock mounted mounting bracket assembly. The mounting bracket shall also serve to further secure the intake dome in place. The motor shall be suitable for continuous duty at the loads it will be subjected to in use.

3.2.4 Variable transformer. A variable transformer shall be provided to regulate the pressure by controlling the motor speed. The variable transformer shall be capable of varying the voltage to the motor from 0 volts to 120 volts when operated from a 120 volt, 60 Hz electric power source. The transformer shall be rated at a minimum of 2.25 amperes and 270 volt-amperes at 120 volts, 60 Hz.

3.2.5 Switch and indicator light. The unit shall be provided with an on-off switch and indicator light. The indicator light, when lit, shall indicate that the power to the unit is on. The switch shall have a minimum rating of 1/3 horsepower at 120 volts 60 Hz. The indicator light shall be of the neon type with an average operation life of 15,000 hours minimum. In lieu of a separate switch and light, a switch with a self-contained lamp that lights when the switch is in the on position shall be acceptable.

3.2.6 Vacuum gage. A vacuum gage of the circular dial type shall be provided with the unit. The gage shall accurately indicate negative pressure from 0 to 60 centimeters of water (cm H_2O). The gage shall be accurate to within $\pm$ 3 percent of reading.

3.2.7 Power cord and plug. A three conductor line cord with three prong male plug conforming to American National Standards Institute (ANSI) Standard C 73.11 shall be provided. The line cord shall be a minimum of 7 feet in free length. Plug shall conform with Underwriters' Laboratories Inc. requirements for Hospital Grade. The plug shall be assembled and installed in accordance with all applicable instructions of the original equipment manufacturer.

3.2.8 Collection bottle assembly. Two one gallon capacity glass collection bottles shall be provided with the unit. The primary collection bottle shall be provided with a metric graduate scale. The graduated scale shall be permanently and legibly printed on the bottle. As an alternate, marking shall be printed on an attachment which can be mounted on the bottle. The collection bottles shall have 70 millimeter continuous thread necks and a nominal outside diameter of 6-5/16 inches. The primary bottle shall be equipped with a cap having two connectors to which the drainage tubes from the chest catheters can be joined, and to which the tubes inside the bottle can be joined. A third connector in the cap shall be provided to which the connector tube from the secondary bottle may be joined. The secondary bottle shall be equipped with a cap having connectors to which the tubing from the primary bottle and the tubing to the suction pump shall be joined. All caps and fittings shall be leakproof when assembled.

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3.2.9 Tubing. All tubing necessary to interconnect the bottles to themselves, to the suction pump and to the chest catheters shall be provided. The tubing from bottle to bottle and bottle to pump shall have a nominal inside diameter of 3/4 to 12/16 inch and a nominal wall thickness of 1/8 inch. The tubing from the patient chest catheters to the primary collection bottle shall be a minimum of 4 feet in length and shall have a nominal inside diameter of 3/8 inch and nominal wall thickness of 1/16 inch. The durometer hardness of all tubing shall be Shore A 70 to 80. The tubing shall be clear polyvinyl plastic. The tubes inside the primary bottle shall have a length of 9-1/2 inches, a nominal inside diameter of 3/8 inch, and a nominal wall thickness of 1/16 inch. All tubing and fittings shall be provided in a clean, sanitary, unused condition.

3.2.10 Receptacle and night light. The receptacle shall be of the three-prong type and shall be securely mounted on the bottom of the cabinet. The night light shall be an integral unit consisting of the light housing, bulb, bulb receptacle, on-off switch and two prong plug. The night light shall accept a 7-watt bulb and shall plug into the receptacle on the underside of the cabinet. The night light shall enable hospital personnel to check the fluid level in the collection bottles without the aid of any additional lights.

3.2.11 Electrical system. The electrical components shall be constructed in accordance with the highest standards of workmanship and safety. The internal wiring shall consist of wires of the type or types which are suitable for the particular application, with respect to the temperature, current, and voltage to which the wire is liable to be subjected, with respect to its exposure to oil or grease, and with respect to other conditions of service to which it is to be subjected. When it is necessary to route wires through holes in sheet metal a smooth busing or smooth well rounded surface shall be provided to prevent abraiding of the wire insulation. Where stranded internal wiring is connected to a wire binding screw, the construction shall be such that loose strands of wire will be prevented from contacting other uninsulated live metal parts and dead metal parts. This shall be accomplished by the use of pressure terminal connectors, soldering lugs, crimped evelets, soldering all strands of the wire together, or any other suitable and reliable means. The entire unit shall be capable of withstanding a 1000 volt dielectric test for one minute immediately following operation (See 4.4.2). The strain relief on the power cord shall be capable of withstanding for one minute, without displacement, a pull of 35 pounds applied to the cord, with the connections within the appliance disconnected. The risk current of the suction apparatus shall not exceed the limits for "cord-connected equipment likely to contact the patient" as specified in the AAMI Safety Standard for Electromedical Apparatus: Safe Current Limits, when measured in accordance with the procedure contained therein before and after the extended operation test described in 4.4.3. A significant increase in risk current after the extended operation (including dielectric withstand) test shall be cause for further investigation of the electrical safety characteristics of the unit.

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3.3 Performance.

3.3.1 Operation. The unit shall operate on 120 volt, 60 Hz, AC electric power. When the on-off switch is in the on position the neon indicator light shall be lit indicating that the unit is operating. The vacuum at the drainage tube connectors shall be continuously adjustable from 0 to 60 cm H_2O. The unit shall be capable of removing 50 liters of air per minute through the system when set at minus 10 cm H_2O and 70 liters of air per minute when set at minus 20 cm H_2O. The vacuum gage shall indicate pump vacuum at all times during operation. The unit shall be capable of aspirating liquids which may accumulate in the pleural space. The unit shall be capable of operating continuously for 168 hours without breakdown (see 4.4.3). Temperature rise during the aforementioned operation shall not exceed 60 degrees Centigrade (60°C.) for the motor; temperature rise of other electrical components shall not exceed 45° C. The vacuum at the patient tubes shall not be less than 90 percent of the vacuum at the pump. The vacuum indicated on the gage shall not increase by more than 10 cm H_2O when the patient tubes are completely blocked (see 4.4.1).

3.4 Finish. Corrosion-resisting steel tubing and sheet shall have a No. 4 finish or better. Carbon steel shall be thoroughly cleaned and degreased and then painted or chromium over nickel over copper plated. Surfaces to be painted shall be given a zinc chromate primer coat followed by two coats of baked-on enamel. Aluminum shall be anodized in accordance with MIL-A-8625, Type II (Sulfuric Acid bath). As an alternate, the aluminum shall be anodized in accordance with MIL-A-8625, Type I or II, or given a chromate chemical film in accordance with MIL-C-5541 and then finished with hammertone enamel. Color shall be such as to provide a pleasing appearance in patient care areas.

3.5 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have

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furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.6 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal and/or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean shall be satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Fusion through the full thickness of the base metal, in areas other than those where 100% penetration is a requirement of joint design, is not acceptable when welding materials over 0.05 inch in thickness. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks which affect the quality.

3.7 Identification marking. A metal plate, of appropriate size, shall be securely attached to each cabinet. The plate shall bear the following information:

Name or registered trademark and Model number of the manufacturer.
Contract or purchase order number.
National stock number.
U. S.

Marking shall be distinct, legible, and of a permanent nature. Marking secured by adhesives shall not be used.

3.8 Workmanship. Workmanship shall be first class throughout. Units shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the components or materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

Table I Sampling for Examination

	Inspection level	AQL (defects per hundred units
For visual examination		
Major defect	S-4	1.0
Minor defects	S-4	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be 100% except that only one unit from each lot shall be subjected to the extended operation test (4.4.3) and risk current test (4.4.4). Failure of extended operation test shall be cause for rejection of the entire lot.

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4.3 Examination. The units shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in Table II.

Table II Classification of Defects

Categories	Defects*
Major	
101	Any component not furnished.
102	Any component not free of damage.
103	Warning not printed on cabinet.
Minor	
201	Finish not as specified.
202	Exposed surfaces of unit not free of burs and sharp edges.
203	Tubing not as specified.
204	Internal wire not secured.
205	Power cord less than 7 feet long.

*Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. The following tests shall be performed to determine compliance with specifications. Test voltage shall be maintained within ± 6 volts during performance tests.

4.4.1 Operation test. The unit shall be completely assembled with bottles and tubes in place and connected to a 120 volt, 60 Hertz power source. The patient tubes shall be connected to an accurate standard calibrated vacuum gage and the variable transformer varied through the entire range of available pressures. The unit shall also be operated at -10 and -20 cm H_2O and the volume of air aspirated per minute shall be determined for compliance with specifications (3.3.1). With the unit operating successively at -20 and -60 cm H_2O the patient tubes shall be completely blocked, the vacuum increase shall be noted. During operation the bottle caps, fittings and tubing shall be examined for air leaks.

4.4.2 Dielectric test. Immediately following the operation test the unit shall be subjected to a dielectric withstand test of 1000 volts between current carrying parts and dead metal parts. The potential shall be slowly raised from 0 to 1000 volts and maintained at that value for one minute. The power switch shall be in the "on" position when test is conducted.

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4.4.3 Extended operation. The unit shall be completely assembled with bottles and tubes and connected to a 120 volt 60 Hertz electric power source. The unit shall then be operated at 30 cm H_2O for a period of 168 hours. The unit shall then be examined for excessive noise and overheating of electrical components (3.3.1). Immediately following the test the unit shall be subjected to the dielectric withstand test specified in 4.4.2. The vacuum being drawn by the unit shall be noted periodically to assure proper operation.

4.4.4 Risk current test. The risk current test shall be conducted before beginning the extended operation test, and again after performing the dielectric withstand test. The test procedures and test equipment used shall comply with all applicable requirements of the AAMI Standard for Electromedical Apparatus: Safe Current Limits. Risk current measurements shall not exceed a limit of 100 microamperes as frequency-weighted by the AAMI Standard Test Load.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One suction apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A. Casters shall be removed and wrapped in kraft paper. Tubing capable of being coiled shall be coiled in such a manner as to prevent kinking or taking a permanent set. The coiled tubing along with any other miscellaneous parts shall be sealed in individual plastic bags and packaged along with the casters in a corrugated fiberboard box of appropriate size and design. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to prevent shifting of or damage to the contents. The collection bottles shall be individually packaged in fiberboard boxes of appropraite size and design. Closure of boxes shall be adequate to prevent spilling of contents. The unit along with the packaged components shall be further packaged in a weather-resistant fiberboard box having a bursting strength of not less than 275 pounds and constructed in accordance with PPP-B-636, grade V3c. The unit shall be so packaged that its weight shall be distributed over the entire surface of the bottom of the box and not carried directly on its legs. Closure and waterproofing shall be as specified in the appendix of the box specification.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be overboxed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, Class 3, Style 3; PPP-B-601, overseas type; PPP-B-621, Class 2; or PPP-B-640, Class 2, Style E. Closure and strapping shall be as specified in the appendix of the applicable box specification. Closure of fiberboard box shall be in accordance with the appendix of PPP-B-640, except that the use of staples, steel and non-metal strapping shall not be used.

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be overboxed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, Class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1 or PPP-B-640, class 1, style E. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard box shall be in accordance with the appendix of PPP-B-640, except that the use of staples, steel or non-metal strapping shall not be used.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Interior packages shall be marked to identify contents and quantity contained.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This suction apparatus is intended for use in medical treatment facilities for maintaining negativity in the pleural space following thoracic surgery.

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Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

This specification covers the following item appearing in the Supply Catalog:

National Stock Number	Item Identification
6515-00-168-6832	SUCTION APPARATUS, SURGICAL, High Volume, Low Pressure, Mobile

This specification does not cover all types, classes, grades or the commodity indicated by the title of this specification, or those commercially available, but is intended to cover the types which ally procured to meet requirements.

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MD
MS
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Preparing activity:
Defense Supply Agency - DSA-DM

Project No. 6515-1778
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